

HD 338492 — Confirmation Checklist

Check	Result	Detail
C1: Gaia quality	P	RUWE=2.05, σ =249.1
C2: Orbit solution	P	P=44.41d, e=0.007, K_1 =80.4
C3: Mass function	P	$f(M)$ =2.39 > Chandrasekhar
C4: Spectral type	P	B9 → T_{eff} =10500 K
C5: HR diagram	P	M_1 =4.40 on MS
C6: Extinction	O	A_V =2.85 (SpT-dependent)
C7: SED fit	O	χ^2 =14.1 (BB approx)
C8: Companion light	P	419× exclusion
C9: X-ray	O	Non-detection (dormant)
C10: UV excess	O	No GALEX coverage
C11: Mass posterior	P	$P(\text{BH})$ =98.5%
C12: Alt. scenarios	F	2/6 not fully excluded
C13: Independent RV	F	Not yet obtained

Score: 7P / 4O / 2F — Verdict: COMPELLING BH CANDIDATE